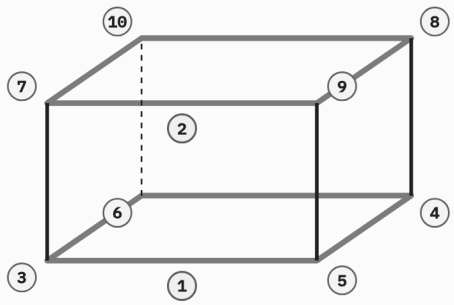
Frame soldering

Ten joints turn two loops and four posts into one frame. About an hour and a half, then a wash. Everything here happens on bare brass, with no electronics anywhere near the bench.

Where you are

done	done	now	next
Cut	**Bent**	**Solder**	**Wash and check**
tube and rod pieces sawn and deburred	two loops, two 6 mm pins ready	2 seams, 4 post bottoms, 4 post tops	flux off, FRAME-01



Stage 1 · joints 1–2
the seam of each loop, on its own, before anything else

Stage 2 · joints 3–6
post bottoms on the bottom loop, in diagonal order: front-left, back-right, front-right, back-left

Stage 3 · joints 7–10
post tops on the top loop, same diagonal order

Diagonal order spreads the pull of each joint, so the frame stays square.

Ten joints, three stages. Blue circles are the two loop seams; orange circles are the eight post joints, numbered in the order you solder them.

Your kit

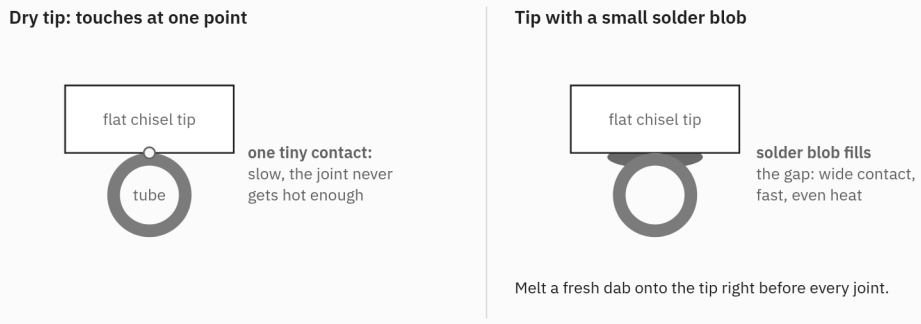
PRODUCT YOU BOUGHT	USE	LINK
Two closed loops · K&S #9831 1.5 mm **TUBE**	top and bottom of the frame	B005WPAW9M
Four posts, filed to 32 mm · K&S #9861 1.0 mm **ROD**	corners	B005WPB7YG
X-Tronic 3020-XTS station, brass-only chisel tip, helping hands, silicone mat	heat, holding, heat-safe surface	owned · no link recorded
Harris SCLF4 Stay-Clean acid flux	brass joints only	B0015DWPV8
MAIYUM 63/37 rosin-core solder, 0.8 mm	every joint	B076QF1Y85
BAISDY wet/dry sandpaper, 400 grit	cleaning each joint shiny	B07D2YYC11
3M Solus 1000 safety glasses	eyes, the whole session	B016KZ1ZPM
NEIKO digital caliper	height checks	B000GSLKIW
Amazon Basics cardstock, a scrap	folded into a 90° square	B0DSH5V1TT
Epic Medical Supply 91 % isopropyl alcohol	drying after the wash	B0BSJ93RKB
Toothpicks, dish soap, two bowls, cotton swabs	applying flux, washing	household
Baking soda	neutralizing the flux	e.g. 365 Baking Soda or any grocery store; skip for now if you have none

> SAFETY, FOR THE WHOLE SESSION
>
> - **Glasses on**, and air moving away from your face: a fan or an open window. Flux fumes sting.
> - **All electronics boxed and closed**, away from the bench. Acid flux must never reach them.
> - **Brass stays hot** for a minute after each joint, and heat travels along the whole loop. Move hot parts with pliers.
> - **Flux on skin:** rinse with water. Keep the bottle closed between joints.
> - **Iron back in its stand** every time you let go of it.

0:00 Set up the iron

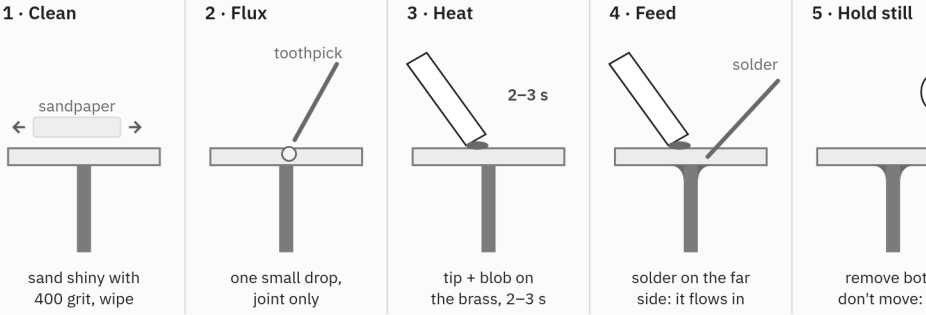
1. **Fit the brass-only tip**, a 2–3 mm chisel, *with the iron cold*. Keep this tip for brass from now on: acid flux wears it out, and you want a clean tip for the electronics later.
2. **Set 370–400 °C (700–750 °F).** Brass soaks up much more heat than a circuit board, so it needs more than electronics work.
3. **Tin the tip:** once it is hot, melt a little solder onto it and wipe it on the damp sponge. It should look silver and shiny.

4 **Lay out the mat**, toothpicks, solder, the flux bottle (closed), the sandpaper and the folded-card square within reach.

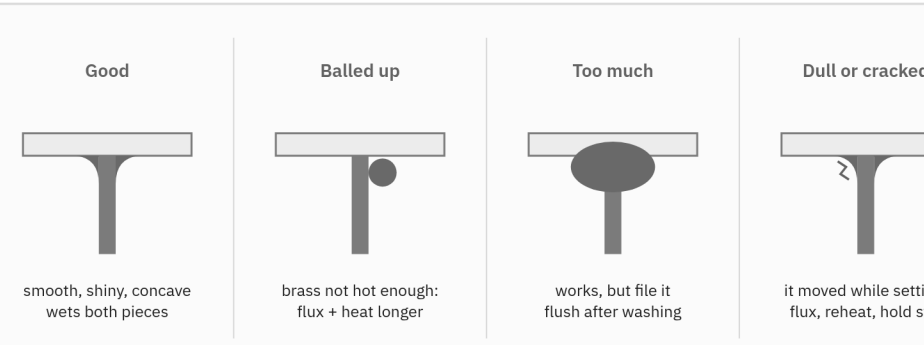


The heat bridge. A flat tip on a round tube only touches along a thin line. A small blob of molten solder on the tip wraps around the tube and carries the heat in. This one habit fixes most "solder will not flow" problems on brass.

How every joint is made



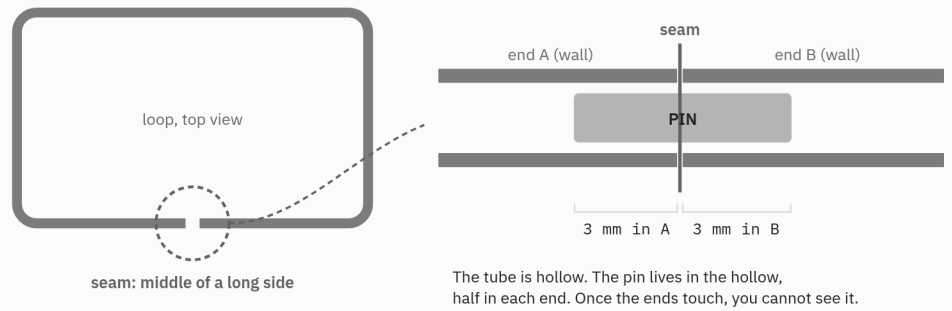
Heat the brass, not the solder. The solder is fed onto the joint on the side away from the iron. If the brass is hot enough it melts the solder itself and pulls it into the gap. Remove the solder wire first, then the iron.



Look at every joint as soon as it sets. Fixing it right away, while the area is still clean and fluxed, is much easier than later.

0:10 The two loop seams

Use: the two **TUBE** loops and the two 6 mm **ROD** pins · K&S #9831 · B005WPAW9M

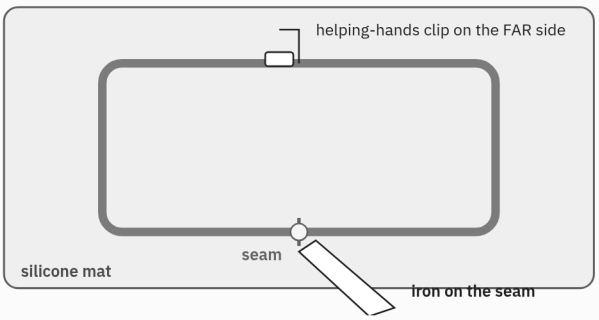


Yes: inside the tube, at the seam. Not in a gap between the ends, and not beside the tube. The two ends close up tight over the middle of the pin, like a peg joining two straws.

PINS IN FIRST, BEFORE ANY HEAT

1. Draw a marker line around the middle of each pin, 3 mm from each end.
2. Spring the loop open a few millimetres at its seam. The loop is springy; it closes again on its own.
3. Push a pin into one tube end up to the line, using the flat, pointed (chain-nose) pliers.
4. Aim the other end straight at the pin and let it slide on until the two ends touch.
5. Check: no gap, no kink at the seam, loop flat. Then solder that seam.

The step-by-step drawings are on the Brass Frame Build page, under “Fitting the pin, step by step”. If a pin will not go in, clear the tube's end with a twist of sandpaper; if it still will not, solder the ends without it.



Why the far side?
A metal clip close to the joint pulls the heat away, and the solder will not flow. Clip at least 20 mm away, or let the loop lie free if it stays put.

Flat on the mat, seam toward you. The mat is heat-safe; the table is not. The pin keeps the ends lined up, so the loop needs no other holding.

1. **Clean** about 3 mm either side of the seam with 400 grit until shiny. Wipe off the dust.
2. **Lay the loop flat** on the mat, seam toward you. Clip the far side only if it moves.
3. **Flux:** one small drop on the seam with a toothpick. Close the bottle.

4 **Heat:** fresh solder blob on the tip, then the tip's flat face onto the seam, touching both tube ends. Hold 2–3 seconds.

5 **Feed:** touch the solder wire to the seam on the side away from the iron. About 1–2 mm of wire melts and runs into the seam.

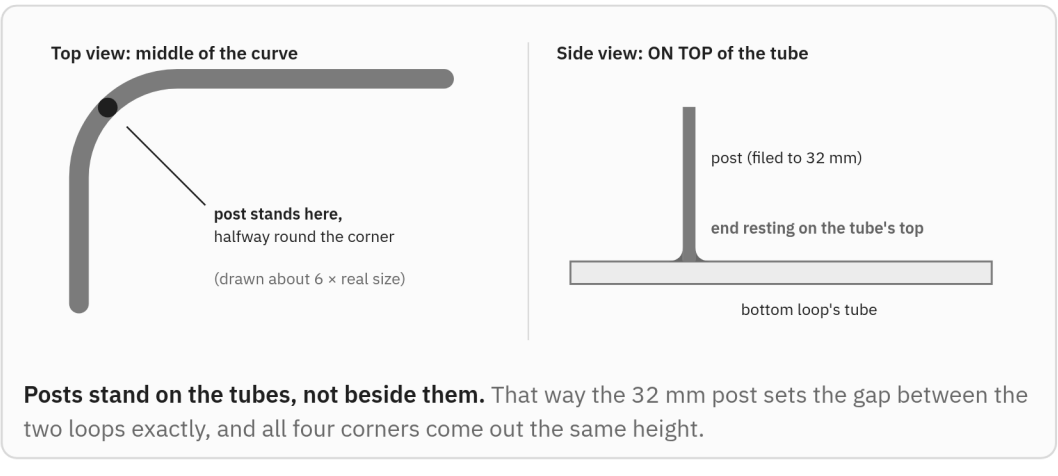
6 **Solder away first, then the iron.** Do not move the loop for 5 seconds.

7 **Check:** a thin, shiny line fills the seam, and the loop is still flat and straight. Then do the second loop.

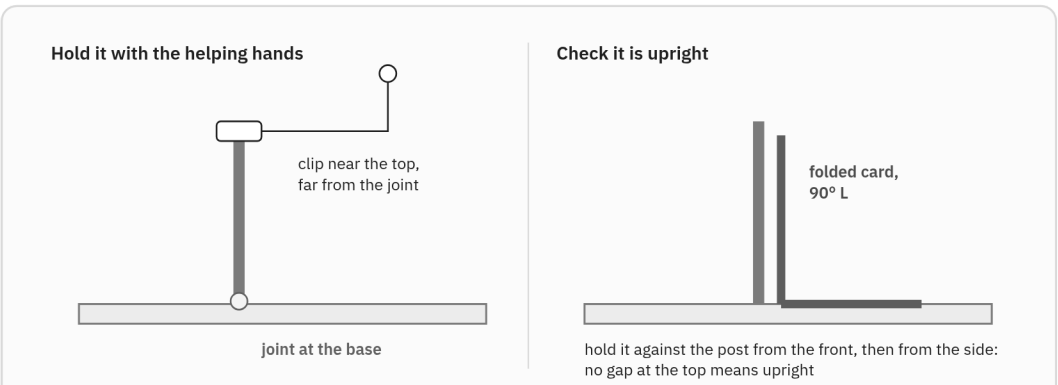
STAGE 2 · JOINTS 3–6

0:25 Posts onto the bottom loop

Use: the four **ROD** posts, 32 mm, solid 1.0 mm · K&S #9861 · B005WPB7YG · plus one soldered loop as the bottom



Posts stand on the tubes, not beside them. That way the 32 mm post sets the gap between the two loops exactly, and all four corners come out the same height.



Fold a scrap of cardstock into an L to make a square. Any card corner is 90°. Check every post in two directions before moving on: a leaning post is easy to fix now and hard later.

1 **Choose the bottom loop** and lay it flat on the mat, seam toward you. The seam side is now the front.

2 **Clean** the middle of each corner curve and both ends of every post with 400 grit.

3 **Post 1, front-left corner (joint 3):** clip the post in the helping hands near its top and stand it on the middle of the corner curve, on top of the tube.

4 **Square it** with the folded card from the front and the side. Adjust the helping hands until there is no gap.

5 **Solder:** drop of flux at the base; tip with blob on the *tube* beside the post, since the tube is the bigger piece and needs most of the heat; solder on the far side of the post. Hold still 5 seconds.

6 **Re-check it is upright.** If it leans: a touch of flux, heat the joint until it is shiny again, nudge the post straight with the clip, and hold still while it sets.

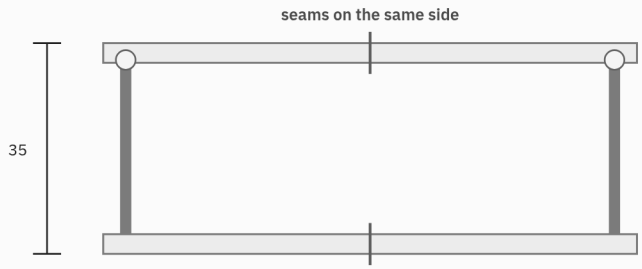
7 **Then the diagonal opposite**: back-right (joint 4), then front-right (joint 5), then back-left (joint 6), squaring and checking each one.

8 **Check all four:** each post top should be the same height, about 33.5 mm above the mat (1.5 mm tube plus the 32 mm post).

STAGE 3 · JOINTS 7–10

0:50 The top loop

Use: the second soldered **TUBE** loop · K&S #9831 · B005WPAW9M



Before soldering the top

· all four post tops touch the top loop

· corners sit straight above the bottom corners

· 35 mm outside at each corner, measured with the calipers

Dry-fit the top loop first. Set it on the post tops, look from the front and from the side, and measure each corner. Only start soldering when all four corners are right.

1 **Clean** the middle of each corner curve on the top loop, and the four post tops.

2 **Set the top loop** on the post tops with its seam on the same side as the bottom seam.

3 **Dry-fit check:** every post top touches the loop at the middle of its corner; each corner sits straight above the one below; the calipers read 35 mm outside at each corner.

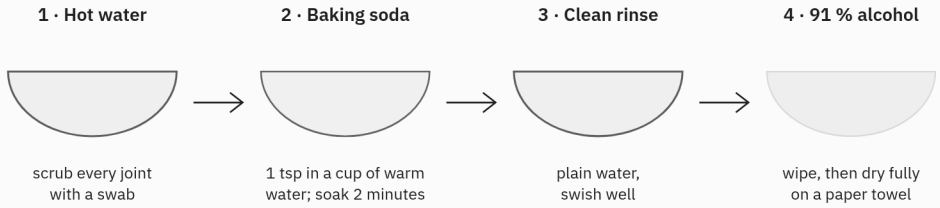
4 **Hold it:** the loop usually stays put by its own weight. If it slides, clip it at the middle of a long side.

5 **Solder front-left (joint 7)**: flux, tip with blob on the loop's tube beside the post, solder on the far side, hold still.

6 **Re-check the height and squareness**, then back-right (8), front-right (9) and back-left (10).

7 **Work quickly.** Heat runs down the post toward its bottom joint. Two or three seconds per joint is plenty; if a bottom joint goes shiny, stop and hold still until it sets.

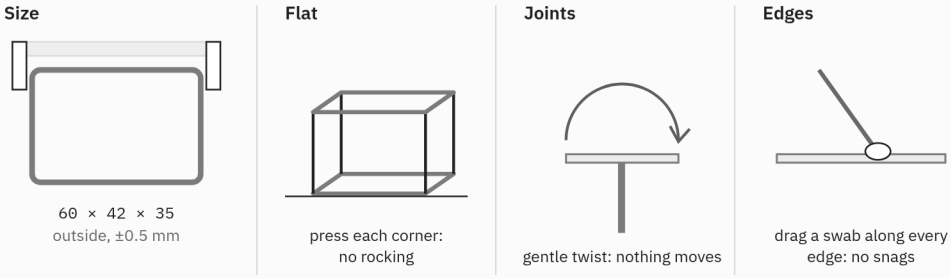
1:20 Wash straight away



Wash straight after the last joint. Leftover zinc-chloride flux corrodes the joints and blisters paint later. If a joint still looks white or crusty after the wash, repeat steps 1 and 2.

No baking soda yet? Scrub every joint in hot water with a drop of dish soap, rinse two or three times, then dry with the alcohol. Do the baking-soda soak when you have it, before paint or any electronics. Never leave flux on overnight.

1:35 Check the frame



A snag on a swab is a burr that would later cut wire insulation. Sand it until the swab slides over it cleanly.

- [ ] **FRAME-01 · THE EMPTY FRAME IS DONE**
  Outside 60 × 42 × 35 mm within about 0.5 mm, sits flat, all ten joints hold a gentle twist, no snags, washed and dry. Send Claude the three outside measurements.

When something goes wrong

WHAT YOU SEE	WHY	FIX
Solder balls up and rolls off	brass not hot enough, or dirty	sand shiny, fresh flux, solder blob on the tip, hold the iron longer
Solder will not stick at all	oxide on the brass	sand again with 400 grit until shiny, then flux
Solder runs along the tube	too much flux or too long on the heat	less flux; small runs file off after washing
A finished joint lets go while soldering the next	heat travelled along the brass	work faster at the new joint; let the frame cool between joints
Post leans after soldering	it moved while the solder set	flux, reheat until shiny, straighten with the clip, hold still
A post does not reach the top loop	post a little short	a gap up to about 0.3 mm fills with solder; bigger: cut and file a new post
Tip turns black and will not take solder	acid flux burned off its coating	wipe on the damp sponge, re-tin straight away; this is why it is the brass-only tip
Flux spits and bubbles	normal: the water in it boiling	use less; keep your face back

After the last joint

- **Tip care:** wipe it, re-tin it, and switch the iron off.
- **Leftover solder bumps** get filed flush with fine sandpaper after the wash.
- **Send Claude** the three outside measurements and a photo of the frame from the front and the side. Next comes the insulation and mounting page.

Sources. Joint order, the frame size and the 32 mm posts follow the Brass Frame Build page and `edu/pages/frame-template-letter.pdf`. Products and the acid-flux washing rule come from `docs/INVENTORY.md`; product links are the original order-sheet listings for the brands the inventory records as bought. The 370–400 °C setting and the order of hot work follow the earlier frame notes in `docs/BUILD_GUIDE.md`.

What is and is not verified. These are standard brass-soldering practices, not measurements from your frame. The checkbox is remembered only in this browser.